

Decision and Risk Analysis

Review of Lectures 1-5, Study Guide for Exam 1

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The event and the action are different

Let $H = 1$ mean an article is fake.

Let a_+ mean *flag it*, and a_- mean *let it pass*.

	$H = 1$: fake	$H = 0$: real
a_+ : flag	correct	false positive
a_- : let pass	false negative	correct

A false positive is a real article flagged as fake.

A false negative is a fake article allowed through.

Which probability enters the decision?

After observing headline information E , define

$$q = P(H = 1 \mid E) = P(\text{fake} \mid E).$$

If we flag the article, the conditional probability that this *chosen action* is wrong is $P(H = 0 \mid E) = 1 - q$.

If we let it pass, the conditional probability that this *chosen action* is wrong is $P(H = 1 \mid E) = q$.

These are conditional probabilities of error for each possible action.

They are not the overall false-positive rate $P(a_+ \mid H = 0)$ or false-negative rate $P(a_- \mid H = 1)$ of a deployed classifier.

Fake-news example: observed counts

In a sample of 150 articles, the headline and truth counts are:

Headline	Fake	Real	Total
Exclamation point	16	2	18
No exclamation point	44	88	132
Total	60	90	150

For a headline with an exclamation point, the empirical conditional probability is $q_I = 16/18 \approx 0.889$.

For a headline without one, $q_{\neg I} = 44/132 = 1/3$.

Example: flag or let pass?

For an exclamation-point headline, $q_! = 16/18$.

Possible action	Error	Probability given “!”
Flag	false positive	$1 - q_! = 2/18 \approx 0.111$
Let pass	false negative	$q_! = 16/18 \approx 0.889$

The two rows describe *alternative actions on the same article*.

The loss attached to each error determines the action; probability alone does not specify it.

Both errors are possible in each headline group

The headline tells us which probability q to use.

The action tells us which mistake could occur.

Headline	Flag: false positive	Let pass: false negative
Has “!”	$1 - q_! = 2/18 \approx 0.111$	$q_! = 16/18 \approx 0.889$
Has no “!”	$1 - q_{\neg!} = 88/132 \approx 0.667$	$q_{\neg!} = 44/132 \approx 0.333$

Each row compares *two possible actions* for one headline group.

For one article, we choose only one action, so we cannot make both mistakes at once.

The entries are conditional chances of being wrong under the action named in that column.

Data, posterior, and prediction

- ▶ The counts 16 fake and 2 real are *data* for the “!” group.
- ▶ 16/18 is an *empirical estimate* of $P(\text{fake} \mid !)$.
- ▶ With an illustrative prior $\theta_! \sim \text{Beta}(1, 1)$, the posterior for the group's unknown fake rate is

$$\theta_! \mid D \sim \text{Beta}(17, 3).$$

- ▶ For the *next* article in that group, the posterior predictive probability is $P(Y_{\text{new}} = 1 \mid D, !) = E(\theta_! \mid D) = 17/20 = 0.85$.

The group-level question $P(\theta_! > 0.5 \mid D)$ concerns the unknown *rate*.

It is a different probability from the next-article prediction.

Costs turn probability into an action

Let C_{FP} be the cost of flagging a real article, and C_{FN} the cost of letting a fake article pass. Correct actions have zero loss in this simple model.

$$R(a_+) = C_{FP}(1 - q), \quad R(a_-) = C_{FN}q.$$

Flag when $R(a_+) < R(a_-)$, equivalently when

$$q > \tau, \quad \tau = \frac{C_{FP}}{C_{FP} + C_{FN}}.$$

For example, if $C_{FN} = 3C_{FP}$, then $\tau = 0.25$.

The empirical $q_I = 0.889$ and $q_{\neg I} = 0.333$ both exceed 0.25.

Check your understanding

Suppose the costs are equal, so $\tau = 0.5$. Use the empirical probabilities.

1. What is the chance of a false positive if we flag a headline without an exclamation point?
2. What is the chance of a false negative if we let it pass?
3. Which action minimizes expected loss for that headline group?

Check your understanding

Suppose the costs are equal, so $\tau = 0.5$. Use the empirical probabilities.

1. What is the chance of a false positive if we flag a headline without an exclamation point?
2. What is the chance of a false negative if we let it pass?
3. Which action minimizes expected loss for that headline group?

Answers: $1 - q_{\neg!} = 2/3$; $q_{\neg!} = 1/3$; let it pass, because $q_{\neg!} = 1/3 < \tau = 1/2$.

Always name the conditioning information, the proposed action, and the consequence before comparing probabilities.

Study guide: updating beliefs

Discrete Bayes rule. For possible states L_i and evidence R ,

$$P(L_i | R) = \frac{P(R | L_i)P(L_i)}{\sum_j P(R | L_j)P(L_j)}.$$

The numerator is prior $\times$ likelihood; the denominator makes posterior probabilities sum to one. If possibilities are eliminated, revise the prior over those that remain before using new evidence.

DAG. Draw an arrow in the model's dependency direction. A prior describes the state before evidence; the likelihood describes how the observation depends on the state; the posterior updates the state.

Beta–Binomial update. If $\theta \sim \text{Beta}(\alpha, \beta)$ and x of n trials are positive, then $\theta | D \sim \text{Beta}(\alpha + x, \beta + n - x)$.

Study guide: from a posterior to a decision

Choose the event that matches the question.

$$q_{\text{rate}} = P(\theta > c \mid D) = 1 - F_{\text{Beta}(\alpha+x, \beta+n-x)}(c)$$

measures support for a state defined by the cutoff c .

For one new binary outcome,

$$q_{\text{next}} = P(Y_{\text{new}} = 1 \mid D) = E(\theta \mid D) = \frac{\alpha + x}{\alpha + \beta + n}.$$

Compare expected losses. With zero loss for correct actions,

$$R(a_+) = C_{\text{FP}}(1 - q), \quad R(a_-) = C_{\text{FN}}q, \quad q > \tau = \frac{C_{\text{FP}}}{C_{\text{FP}} + C_{\text{FN}}} \Rightarrow a_+.$$

At equality, either action has the same expected loss.

The state cutoff c and the action threshold τ serve different purposes.

Exam instructions

Closed-book exam.

You may use pen or pencil,
a basic or scientific calculator, and
one regular sticky note (standard 3x3 *Post-it* or equivalent).

Leave the sticky note attached to your exam when you turn it in.

No other materials or devices are permitted
(NO smartphones, tables, computers, smart-glasses, etc.)

Do your best, always. I believe in you.

Have a blessed weekend, see you all on Monday! ッ